INCH-POUND

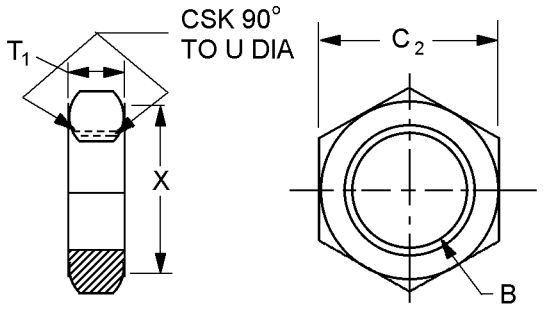
MS51860B
w/AMENDMENT 1
30 January 2020
SUPERSEDING
MS51860B
25 July 2018

DETAIL SPECIFICATION SHEET

# LOCKNUT, BULKHEAD, 37 DEGREES FLARED AND FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.




See notes at end of figure.

FIGURE 1. Locknut bulkhead.

AMSC N/A

FSC 4730

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| Dash number<br>Style II<br>(see note 7)/ | Tube OD Nom inches (mm) | B Straight thread Class 2B | $C_2$ Hex Nominal | $T_1$ inches (mm) ±.02 (0.5) |
|---|---|---|---|---|
| -51 | .1250 (3.175) | .3125 - 24UNF | 9/16 | .22 (5.6) |
| -52 | .1875 (4.763) | .3750 - 24UNF | 5/8 | .22 (5.6) |
| -53 | .2500 (6.350) | .4375 - 20UNF | 11/16 | .25 (6.4) |
| -54 | .3125 (7.936) | .5000 - 20UNF | 3/4 | .25 (6.4) |
| -55 | .3750 (9.525) | .5625 - 18UNF | 13/16 | .27 (6.9) |
| -56 | .5000 (12.700) | .7500 - 16UNF | 1 | .31 (7.9) |
| -57 | .6250 (15.875) | .8750 - 14UNF | 1-1/8 | .36 (9.1) |
| -58 | .7500 (19.050) | 1.0625 - 12UN | 1-3/8 | .41 (10.4) |
| -59 | .8750 (22.225) | 1.1875 - 12UN | 1-1/2 | .41 (10.4) |
| -60 | 1.0000 (25.400) | 1.3125 - 12UN | 1-5/8 | .41 (10.4) |
| -11 | 1.2500 (31.750) | 1.6250 - 12UN | 1-7/8 | .41 (10.4) |
| -12 | 1.5000 (38.100) | 1.8750 - 12UN | 2-1/8 | .41 (10.4) |
| -13 | 2.0000 (50.800) | 2.5000 - 12UN | 2-3/4 | .41 (10.4) |

| Dash number<br>Style II<br>(see note 7) | U Diameter +.015 (0.38) -.000 | X Diameter +.005 (0.13) |
|---|---|---|
| -51 | .317 (8.05) | .562 (14.27) |
| -52 | .380 (9.65) | .625 (15.88) |
| -53 | .443 (11.25) | .688 (17.48) |
| -54 | .505 (12.83) | .750 (19.05) |
| -55 | .567 (14.40) | .812 (20.62) |
| -56 | .755 (19.18) | 1.000 (25.40) |
| -57 | .880 (22.35) | 1.125 (28.58) |
| -58 | 1.067 (27.10) | 1.375 (34.93) |
| -59 | 1.193 (30.30) | 1.500 (38.10) |
| -60 | 1.317 (33.45) | 1.625 (41.28) |
| -11 | 1.630 (41.40) | 1.875 (47.63) |
| -12 | 1.880 (47.75) | 2.125 (53.98) |
| -13 | 2.050 (52.07) | 2.750 (69.85) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Threads shall be in accordance with ASME B1.1.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
7. Style I dash numbers cancelled after 22 November 1974. Dimensions for style I locknuts see design standard MS51888.

FIGURE 1. Locknut, bulkhead - Continued.



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REQUIREMENTS:

Locknut shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be as specified in table I.

All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The locknut shall show no evidence of red corrosion after 96 hours of salt spray. Internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.



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Table I. Material and finish designators.

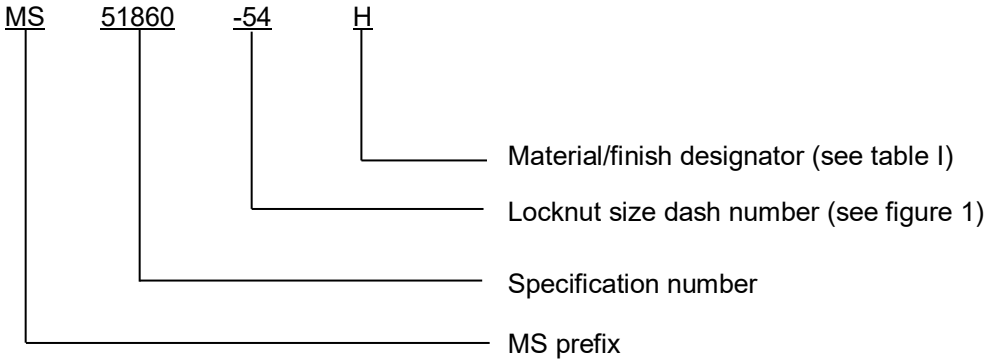
| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| Blank | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 2 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 7/ |
| H | Steel | Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/ |
| J | Steel | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/ |
| M | Nickel-copper alloy UNS N04400 | No additional finish. |
| N | High-chromium nickel alloy UNS N06690 | No additional finish. |
| P | Steel | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/ |
| R | Steel | Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/ |
| SS | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4. |
| T | Titanium | Anodize in accordance with SAE-AMS2488 type 2. 6/ 8/ |
| TF | Titanium | Fluoride phosphate in accordance with SAE-AMS2486. 6/ 8/ |
| Z | Steel | Zinc plating in accordance with ASTM B633, type VI, Fe/Zn 8. 4/ |
| ZC | Steel | Zinc may be any zinc plating from PIN codes J and R with a colored chromate coating. 4/ 5/ |

1/ Embrittlement test need not be run.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army ground systems. For new designs PIN codes Blank, R, and ZC are not to be supplied to Army and Marine ground forces.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.
7/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA. The use of cadmium plating for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.
8/ Titanium shall not be used in oxygen or potable water systems.



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PIN: The PIN consists of the letters “MS”, the specification number, a dash number for the locknut size, and a letter for material finish designator.




PIN example: MS51860-54H indicates a locknut, bulkhead, .6250 inch (15.875 mm), steel with zinc-aluminum plating.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

The use of cadmium plating for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.

For new designs PIN codes Blank, R, and ZC are not to be supplied to Army and Marine ground forces.

Class I and II ozone depleting substances (ODS) shall not be used in MS51860 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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Referenced documents. In addition to MIL-DTL-18866, this document references the following:

| | | |
|---|---|---|
| MIL-DTL-16232 | SAE-AMS-STD-595/36081 | SAE-AMS-STD-595/36270 |
| MS51888 | SAE-AMS-STD-595/36099 | SAE-AMS-STD-595/36280 |
| ASME B1.1 | SAE-AMS-STD-595/36118 | SAE-AMS-STD-595/36293 |
| ASTM B117 | SAE-AMS-STD-595/36134 | SAE-AMS2417 |
| ASTM B633 | SAE-AMS-STD-595/36152 | SAE-AMS2486 |
| ASTM F1136/F1136M | SAE-AMS-STD-595/36170 | SAE-AMS2488 |
| ASTM MNL37 | SAE-AMS-STD-595/36173 | SAE-AMS2700 |
| SAE-AMS-C-81562 | SAE-AMS-STD-595/36176 | SAE-J514 |
| SAE-AMS-QQ-P-416 | SAE-AMS-STD-595/36231 | T9070-AL-DPC-020/077-2 |
| SAE-AMS-STD-595/36076 | SAE-AMS-STD-595/36251 | |

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2020-022)

Review activities:
Army - AT, GL, MI
Navy - AS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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